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Palaeomagnetism of Precambrian basic intrusion and dykes of Northern Karelia, eastern Fennoscandian Shield

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Abstract

Palaeomagnetic results for mafic and ultramafic intrusions and their country rocks in Russian Karelia are presented. The oldest rock type is pyroxene–hornblende gabbro, regarded as late Archaean or Palaeoproterozoic in age. Later, at the early stage of formation of a volcanic–plutonic association, including layered intrusions and metavolcanites, gabbro–norite dykes were emplaced. Sm–Nd isotopic analyses of this association yield mineral and whole-rock ages of 2.47–2.35 Ga with $\epsilon_{Nd} = -0.4$ to 1.2 for gabbro–norite dykes. Based on field relations, the youngest rock types are pyroxene–plagioclase porphyry dykes and gabbro–dolerite dykes.

Three distinct magnetization components were isolated by alternating field and thermal demagnetization. The most common remanence component B, with a mean direction of $D = 30°$, $I = 58°$, $k = 52$, $\alpha_{95} = 10°$ ($N = 5$ geological units), is an overprint interpreted to have been acquired at the last stage of the Svecofennian orogeny (1.90–1.70 Ga). Component A, with a mean direction of $D = 350°$, $I = 49°$, $k = 69$, $\alpha_{95} = 9°$ ($N = 5$ geological units), is also an overprint, probably acquired at the onset of the Svecofennian orogeny. Both components are found in all rocks studied. Component C, with a mean direction of $D = 126°$, $I = 73°$, $k = 167$, $\alpha_{95} = 10°$ ($N = 3$ geological units), is isolated as a primary remanence in the pyroxene–hornblende gabbro intrusion and gabbro–norite dykes.

The results are compared with other contemporaneous palaeomagnetic poles from Finland and Russian Karelia.

1. Introduction

The Apparent Polar Wander Path (APWP) for the Fennoscandian Shield has been recently constructed for the time interval 2.85–0.60 Ga (Elming et al., 1993). It comprises palaeomagnetic poles from the Archaean craton, the Svecofennian and the Sveconorwegian provinces. The Archaean (Kola–Karelian) craton consists of the Karelian province which is a typical granite–greenstone belt terrain, and the Belomorian and Kola provinces consisting of high-grade gneisses (Gaál and Gorbatschev, 1987). The Archaean and Palaeoproterozoic palaeomagnetic data from the Fennoscandian Shield are predominantly obtained from the Karelian and Svecofennian provinces. Few poles are from the Lapland granulite belt of the Belomorian province and from the Kola province. The consistency of poles of the same age from different regions of the Karelian and Svecofennian provinces has permitted the construction of the Palaeoproterozoic part of the APWP of the Fennoscandian Shield. No palaeomagnetic evidence has been found for major tectonic rotations or relative dis-

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placements between these parts of the shield. However, conclusions based on the data from the Archaean and Proterozoic are still preliminary due to the scarcity of palaeopoles for these time periods. In this study, the Karelian palaeomagnetic investigations have been continued and combined with high-precision geochronological studies in order to improve the APWP of the Fennoscandian Shield.

2. General geology

The study area is located between the lakes Piaozero and Tikshozero. It is bordered by the Belomorian province (Fig. 1). The bedrock is composed of Archaean granites, granodiorites and migmatites. Basement rocks have undergone metamorphic and structural reworking more than once. There is a gradual transition between all the rock types. A comprehensive description of the geology is given by Shemiakin and Shoorkin (1974), Sidorenko (1974), Lavrov (1979), Ein (1982, 1984) and Stepanov (1984). The Archaean basement is cut by pyroxene–hornblende gabbro intrusions, gabbro–norite dykes, associated with layered intrusions, pyroxene–plagioclase porphyrite and gabbro–dolerite dykes. Sm–Nd ages of the Sumian–Sariolian volcanic–plutonic association to which the layered intrusions and the gabbro–norite dykes of Karelia belong, are 2.47–2.36 Ga (Levchenkov et al., 1990; Amelin and Semenov, 1990). The gabbro–norite dyke of this study has an Sm–Nd age of 2470 ± 0.065 Ga (ϵ_{Nd} −0.4).

Pyroxene–hornblende gabbro intrusions are not dated, but Lavrov (1979) considers them to be of the same age as the layered intrusions, while Ein (1982) suggests that they are of Archaean age. According to Stepanov (1984) there are at least three generations of gabbro–dolerite dykes in the area. Based on cross-cutting relations they are the youngest rocks in the area.

The area has repeatedly metamorphosed up amphibolite-facies grade during the Archaean, but the Palaeoproterozoic metamorphism did not exceed the greenschist facies. The Proterozoic rocks of this study do not show signs of essential alterations and there are no indications of structural tilting.

3. Classification and description of the geological units

3.1. Pyroxene–hornblende gabbro intrusions

The pyroxene–hornblende gabbro intrusions are located in a SE–NW-trending zone about 10 km wide and 30 km long (Fig. 1). They comprise pyroxene–hornblende gabbros, gabbro–norites and anorthosites. These intrusions have an irregular spotty structure, and there is a gradual transition between the different rock

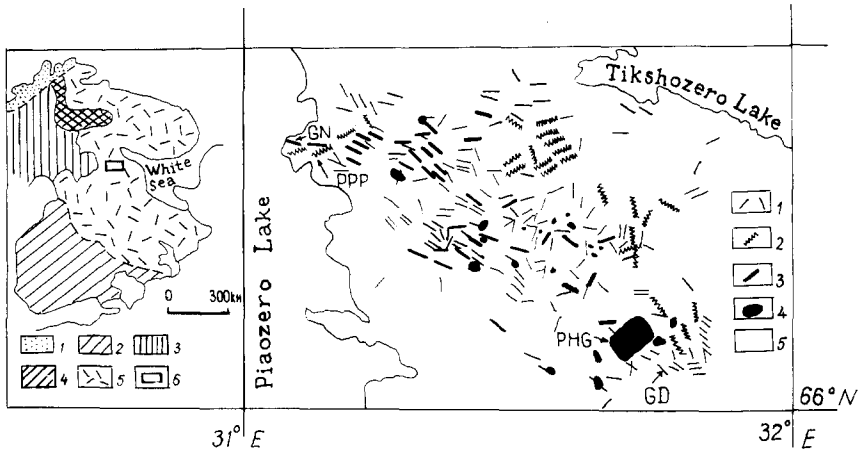


Fig. 1. At the left, general geological map of the Fennoscandian Shield: 1 = Caledonides; 2 = Svecofennides; 3 = Karelides within Archaean basement; 4 = Lapland granulite belt; 5 = Archaean basement; 6 = study area. At the right, location of mafic–ultramafic intrusions and dykes in the study area (after Ein, 1982, 1984): 1 = gabbro–dolerite dykes; 2 = pyroxene–plagioclase porphyrite dykes; 3 = gabbro–norite dykes; 4 = pyroxene–hornblende gabbro intrusions; 5 = Archaean granitoids; *GD*, *PPP*, *GN*, *PHG* = palaeomagnetic sites.

types. The main minerals are plagioclase, hornblende, orto- and clinopyroxene and biotite.

3.2. Gabbro–norite dykes

E–W- to SE–NW-trending gabbro–norite dykes with a steep northerly dip ($>65°$) are widespread in the area (Fig. 1). The width of the dykes is typically less then 60 m. In places the country rocks are melted in the endocontact zones of dykes. The main minerals are olivine, orto- and clinopyroxene and plagioclase.

3.3. Pyroxene–plagioclase porphyry dykes

Pyroxene–plagioclase porphyry dykes follow an arch-shaped fracture zone in the area (Fig. 1). The width of the dykes varies from 2 to 55 m. The marginal parts of the dykes are typically composed of gabbro–dolerite with rare porphyritic grains of pyroxene, plagioclase and sometimes blue quartz. Towards the centre, the number of inclusions increases and the rock passes into porphyrite. The porphyritic zone is generally absent in the smaller dykes.

3.4. Gabbro–dolerite dykes

The gabbro–dolerite dykes represent the youngest rock unit in the area. They are numerous in Northern Karelia. The strike of the dykes varies from E–W to NE–SW, with NW–SE-trending dykes dominating (Fig. 1). The width is typically less than 35 m. The main minerals are plagioclase and pyroxene, in places partly or wholly replaced by hornblende.

The four sampled dykes trend NW–SE, N–S and NE–SW and probably present different dyke generations. Their width is 10, 15, 13 and 0.4 m.

4. Sampling and measurement

Samples were taken as hand samples. In-situ orientation was made with a magnetic compass. Altogether 65 samples were taken from 19 sites: 31 samples were taken at 10 sites of one pyroxene–hornblende gabbro intrusion, 6 samples were taken at one site of one gabbro–norite dyke, 5 samples were taken at two sites in one pyroxene–plagioclase porphyrite dyke, 15 samples were taken at 4 sites of 4 gabbro–dolerite dykes and 8 country-rock samples were taken at 2 sites in the vicinity of the contact of the pyroxene–hornblende gabbro intrusion.

From each sample one or two specimens were obtained. Measurements were made with a static magnetometer (model MA-21). Progressive alternating field (AF) demagnetization (peak up to 100 mT) (Fig. 2) or thermal demagnetization (heating up to 670°C)

(Fig. 3) was used to isolate a characteristic remanent magnetization. The results of AF- and thermal demagnetizations are in good agreement. To recognize the characteristic remanence, Zijderveld diagrams (Zijderveld, 1967) were used for visual inspection of the components. The components were isolated as straight-line segments.

5. Palaeomagnetic results

All the rock types carry two or three remanence components, A, B and C, which were isolated in AF-demagnetization steps of 20–100 mT and at temperatures of 200–670°C. A soft remanence component which forms 15% of the total magnetization is removed in the first AF-demagnetization steps of 0–20 mT or at temperatures of 0–200°C and it probably represents the present Earth's magnetic field.

Most samples yielded two components, but some samples (e.g., gabbro–dolerites) only one. A, B and C components never occur together.

Fisher's (1953) statistics were used to calculate the mean magnetic directions. All magnetic data are based on sample means combined to site means.

The palaeomagnetic data are listed in Table 1 and plotted in Figs. 2–5.

Component A is revealed in 24 specimens and is probably carried by haematite, because it is removed only at high-peak fields (Fig. 2). Component B represents the most common magnetization (Fig. 2), and was isolated in 51 of 65 specimens. Component C is found only in the pyroxene–hornblende gabbro, the gabbro–norite and the country rocks, and was isolated in 10 specimens. Component B is probably carried by both magnetite and haematite (Figs. 2a and 2d) and component C is mainly carried by magnetite (Figs. 2a, 2b and 3).

Because the determination of true polarity in the Precambrian is still ambiguous, we have adhered the

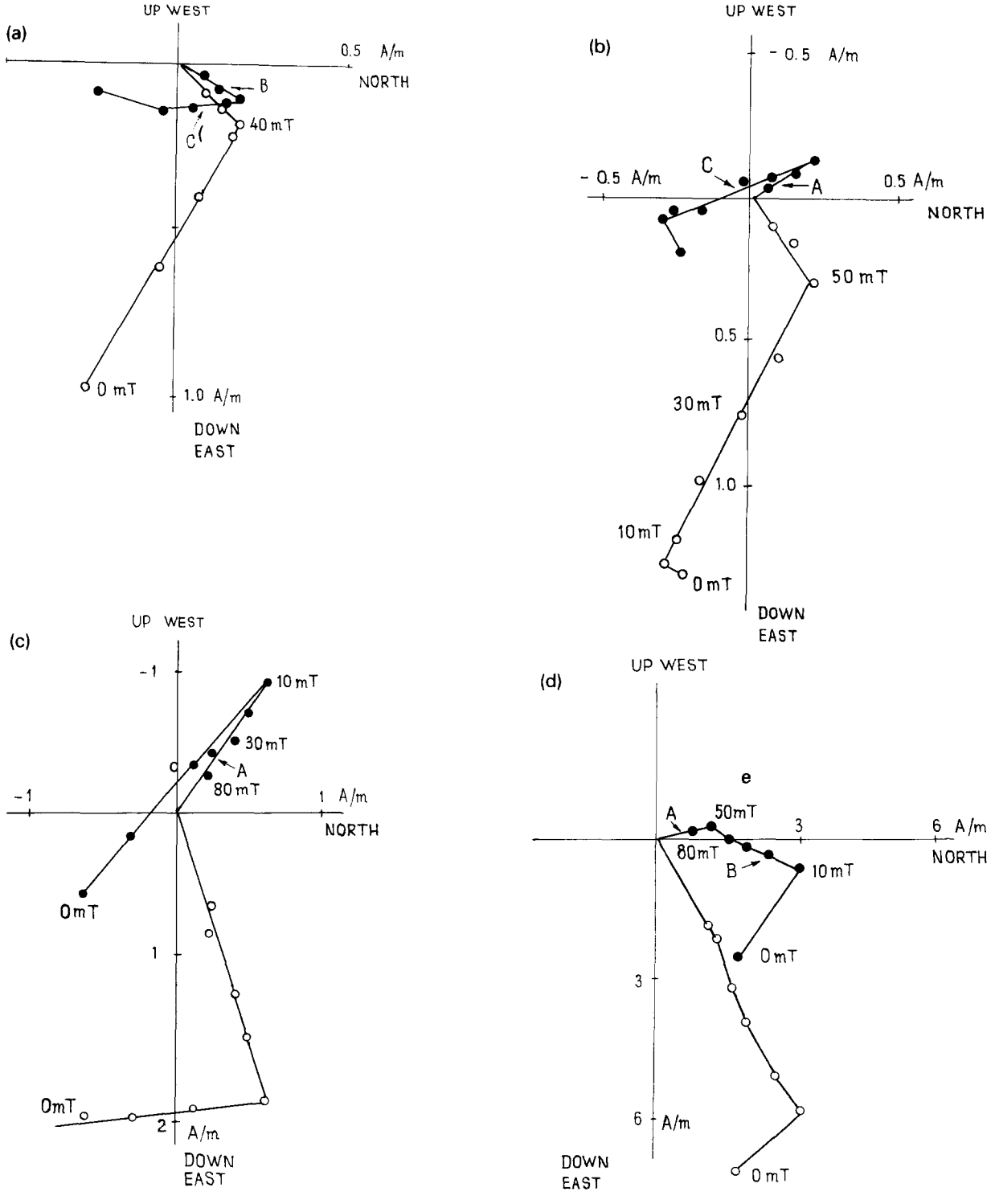


Fig. 2. Orthogonal vector projections (Zijderveld diagrams) showing alternating field demagnetization behaviour for pyroxene–hornblende gabbro (a), gabbro–norite (b), pyroxene–plagioclase porphyrite (c) and gabbro–dolerite (d). Solid (open) symbols denote points in the horizontal (vertical) plane.

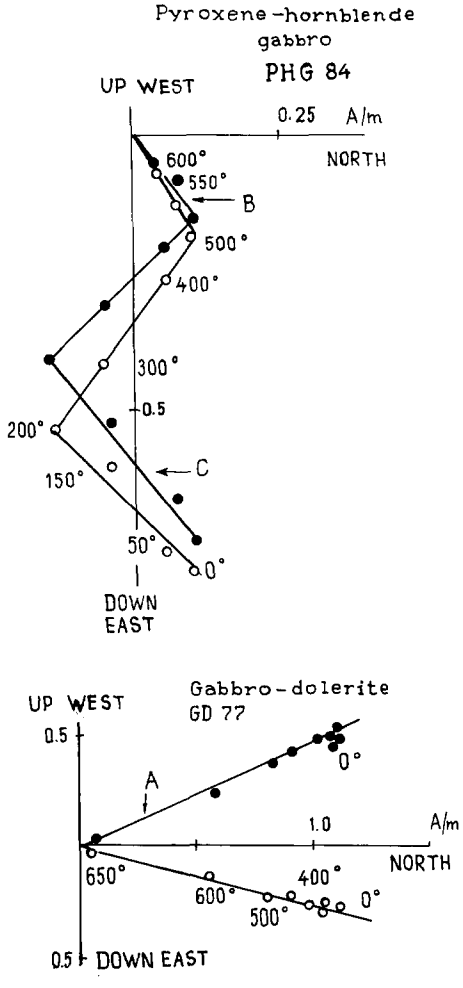


Fig. 3. Orthogonal vector projections showing thermal demagnetization behaviour for pyroxene–hornblende gabbro and gabbro–dolerite. Symbols as in Fig. 2.

definitions of Pesonen and Neuvonen (1981) in this study. The polarity is normal if the north-seeking palaeomagnetic direction yields a pole in the Pacific or in the Canadian Arctic. Otherwise the polarity is reversed. Hence, in agreement with this determination, we consider A and B as normal-polarity components and C as a reversed one. We have not revealed both normal and reversed palaeomagnetic directions in the geological units studied. The directions of the components A, B and C are presented in Table 1 and Fig. 4.

Component C is supposed to be the oldest remanence component because it has been isolated only in the older units. It has not been found in the younger pyroxene–plagioclase porphyrites and gabbro–dolerites. Possible origin for the C remanence is the widespread mafic–ultramafic magmatism at 2.47–2.35 Ga in the area (Levchenkov et al., 1990). Component C is considered primary in the gabbro–norite dykes and as an overprint in the Archaean country rocks and also in the pyroxene–hornblende gabbro if the latter is of Archaean age, as suggested by Ein (1982). If the pyroxene–hornblende gabbro is of Palaeoproterozoic age, as suggested by Lavrov (1979), then the C component in the gabbro can be considered primary.

Components A and B are typical overprints in the pyroxene–hornblende gabbro, in the gabbro–norite dykes and in the country rocks. In these rocks they almost totally overprint component C. In the pyroxene–plagioclase porphyrite and the gabbro–dolerite dykes only components A and B have been found. That means that one is primary, acquired during cooling of the formation, and the other is an overprint. Alternatively, both are secondary overprints. Since, based on cross-cutting relations, the pyroxene–plagioclase porphyrites are older than the gabbro–dolerites, the latter assumption is more probable. Based on the dominance of component B in the rocks, it is suggested that component A is older then B.

The ages of the remanences of the rocks of Northern Karelia can be estimated by comparing the results with the APWP of the Fennoscandian Shield. The A, B and D components (not to be confused with A, B and C of this paper) of Mertanen et al. (1989) from layered intrusions in Northern Finland have nearly the same directions as the A, B and C components of the Northern Karelian rocks of this paper (Fig. 5). The steeper inclinations of the Karelian rocks, compared with the Finnish ones are probably due to imperfect cleaning of viscous components, possibly the present Earth's magnetic field.

The age order of the components in the layered intrusions of northern Finland was shown to be from D to A and to B (Mertanen et al., 1989). The A and B components were supposed to be related to the Svecofennian orogeny, where A represents the early stage at 1.88 Ga and B the later stage of the orogeny at 1.75 Ga. Probably the same geological processes caused the C–A–B order for the remanences of the Northern Karelian rocks.

Table 1
Characteristic remanence components and pole positions of the rocks of northern Karelia

Rock units/samples	Sampling place		$M/N/n$	D (°)	I (°)	α_{95} (°)	k
	ϕ (°N)	λ (°E)					
Component A							
PHG 72, 83, 88, 98, 115–116, 119–120	66.0	31.8	1/6/9	357	45	4	250
GN 128, 129	66.2	31.0	1/1/2	330	47	–	50
PPP121–123	66.2	31.0	1/1/3	350	46	26	23
GD 77, 89–92, 106–108	66.0	31.8	4/4/8	353	47	5	300
CR 101, 105	66.0	31.8	1/1/2	342	54	–	25
Mean for component A			5*/13/24	350	49	9	69
Pole position: lat. = 53°N; long. = 228°E; A_{95} = 10°							
Component B							
PHG 70–77, 79–82, 84–88, 97–100, 113, 115, 117, 118	66.0	31.8	1/10/31	31	59	4	129
GN 126–128, 130	66.2	31.0	1/1/7	26	56	13	22
PPP 124–125	66.2	31.0	1/1/3	47	23	23	28
GD 90, 93, 109, 111	66.0	31.8	3/3/5	21	57	12	100
CR 94–96, 103	66.0	31.8	1/2/5	23	73	–	25
Mean for component B			5*/17/51	30	58	10	52
Pole position: lat. = 57°N; long. = 165°E; A_{95} = 13°							
Component C							
PHG 70, 73, 75, 84, 87	66.0	31.8	1/3/5	124	71	21	35
GN 129, 130	66.2	31.0	1/1/2	126	68	–	4
CR 102, 105	66.0	31.8	1/1/3	128	80	23	30
Mean for component C	3*/3/10	126	73	10	167		
Pole position: lat. = 41°N; long. = 65°E; A_{95} = 10°							

Units: PHG = pyroxene–hornblende gabbro intrusion; GN = gabbro–norite dyke; PPP = pyroxene–plagioclase porphyry dyke; GD = gabbro–dolerite dyke; CR = country rocks. M = number of geological units; N = number of sites; n = number of specimens (* = statistical level used in mean calculation); α_{95} = radius of 95% confidence circle about the mean direction; k = the precision parameter (Fisher, 1953); A_{95} = radius of confidence circle about the mean pole position.

A comparison was also made with palaeomagnetic data from the Burakovskaja layered intrusion and a connected dyke from southeastern Karelia (Krasnova and Gooskova, 1990). In the pyroxenite of the southeastern part of the Burakovskaja intrusion (Aganozero block) and in the connected dyke only one characteristic remanence component was isolated. But, it appeared that the direction of the characteristic component of the Aganozero block pyroxenites differs from that in the dyke. After tectonic correction of the Aganozero block (clockwise rotation of about 70°) the characteristic remanence component of the Burakovsky complex rocks is in good agreement with the C (this work) and D (Mertanen et al., 1989) components of northern Karelia and northern Finland rocks (Fig. 5). Moreover, the difference (about 70°) in the direction of the magnetic field isolines of the Aganozero block and the remainder of the intrusion (demonstrated by Logatschev and Zakharov (1979) and caused, according to them, by the rotation or tilting of the block) disappeared after tectonic correction.

Pole positions for different rock types in northern Karelia are given in Table 1. The Archaean–Palaeoproterozoic part of the latest APWP for the Fennoscandian Shield is presented in Fig. 6. The palaeopoles for the layered intrusions in northern Finland and southeastern Karelia are plotted in the APWP. The poles of northern Karelia are in agreement with the APWP (Fig. 5).

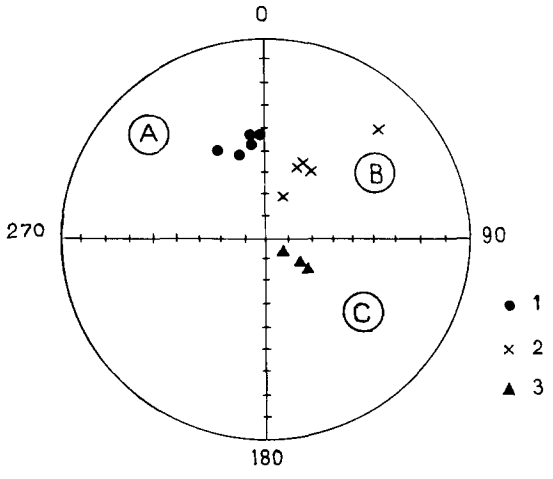


Fig. 4. Mean remanence components for different rock types: 1 = component A, 2 = component B, and 3 = component C.

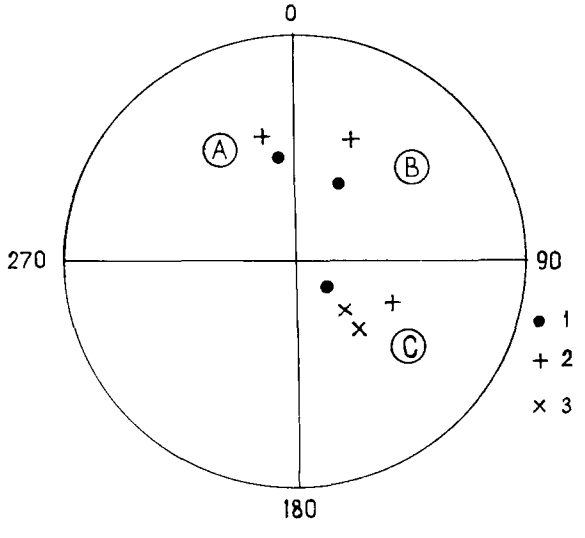


Fig. 5. Mean directions of remanence components: 1 = this study, 2 = layered intrusions of northern Finland, and 3 = Burakovsky complex of Russian Karelia.

Deviations of poles B_1 and C_1–C_3 from the corresponding dated poles of the APWP are probably due to the scatter of data rather than being real. However, the order of palaeopoles from C to A and to B is in accordance with the Palaeoproterozoic part of the APWP of the Fennoscandian Shield.

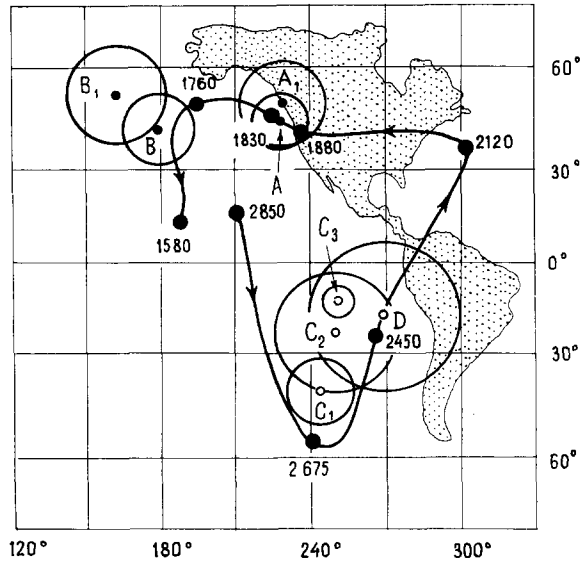


Fig. 6. Palaeomagnetic poles of the Palaeoproterozoic basic intrusions and dykes of the Karelian craton on the APWP of the Fennoscandian Shield (after Elming et al., 1993). Poles: *A*, *B*, *D* after Mertanen et al. (1989), northern Finland; A_1, B_1, C_1 this work, northern Karelia; C_2, C_3 after Krasnova and Gooskova (1990), southeastern Karelia (Burakovsky complex).

6. Conclusions

The Precambrian rocks of northern Karelia, Russia have a complicated magnetization history. Their remanence is multicomponent and was caused by intense mafic–ultramafic magmatism at 2.47–2.35 Ga and by the later Svecofennian activity at 1.90–1.70 Ga of the Karelian craton. During Svecofennian times the rocks acquired overprints twice: in the early and in the late stage of the Svecofennian orogeny. The palaeomagnetic information combined with the known Proterozoic endogenic activity allows us to conclude that the strength of these processes decreased from the early stages onwards. Another conclusion is that some stages of the endogenic activity were accompanied by alterations of the ferromagnetic minerals (e.g. high- and low-temperature oxidation).

The agreement between palaeopoles of similar age from northern and southeastern Karelia and from northern Finland implies that no significant movements have

taken place between the different parts of the Karelian craton since the beginning of the Palaeoproterozoic.

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